# Assignment 6: Rigid Body Dynamics

Robot Kinematics and Dynamics

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Due: See Canvas for the due date

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# 1 Overview

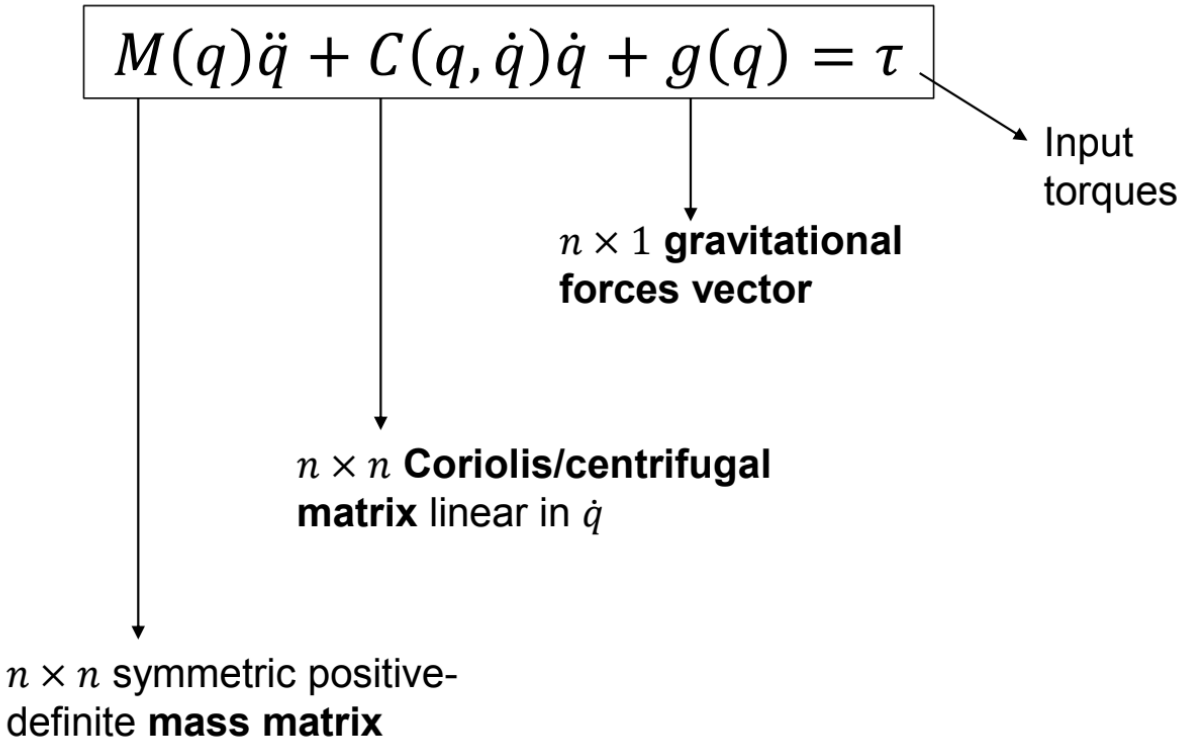
This assignment reinforces the following topics:

- Rigid Body Dynamics

# 2 Background

## 2.1 Standard Form

As a reminder, the standard form for writing equations of motion is as follows:



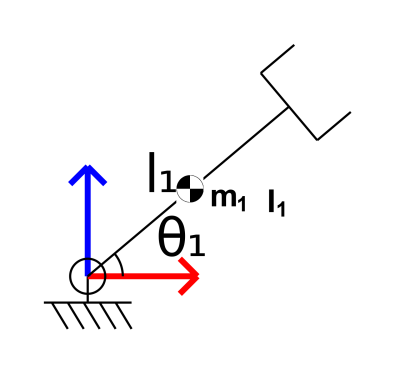
# 3 Instructions

- See Canvas for the due date.
- Submit your written answers to Gradescope as a single PDF named `writeup.pdf`. There is no code submission for this assignment. See the complete submission checklist at the end, to ensure you have everything.
- You are free to write `writeup.pdf` using any method, but we recommend LaTeX for consistent formatting. You can upload the assignment zip to Overleaf and edit `assignment6.tex` directly.
- Each question (for points) is marked with a **points** heading.
- **Start early!** This homework may take a long time to complete.
- **During submission indicate the answer/page correspondence carefully when submitting on Gradescope.** If you skip a written question, just submit a blank page for it. This makes our work much easier to grade.
- If you have any questions or need clarifications, please post in Piazza or visit the TAs during the office hours.
- Unless otherwise specified, **all units are in radians, meters, and seconds**, where appropriate.

# 4 Written Section

1) Rigid Body Dynamics

Please use the diagram of the arm below for the following questions:




The arm has a link length of $l_1$ where the center of mass is located in the center of the link at $\frac{l_1}{2}$. The link's mass is $m_1$ and it has moment of inertia $I_1$.

(1) [5 points] Determine the Kinetic Energy for the R arm shown above.

(2) [5 points] Determine the Potential Energy for the R arm shown above.

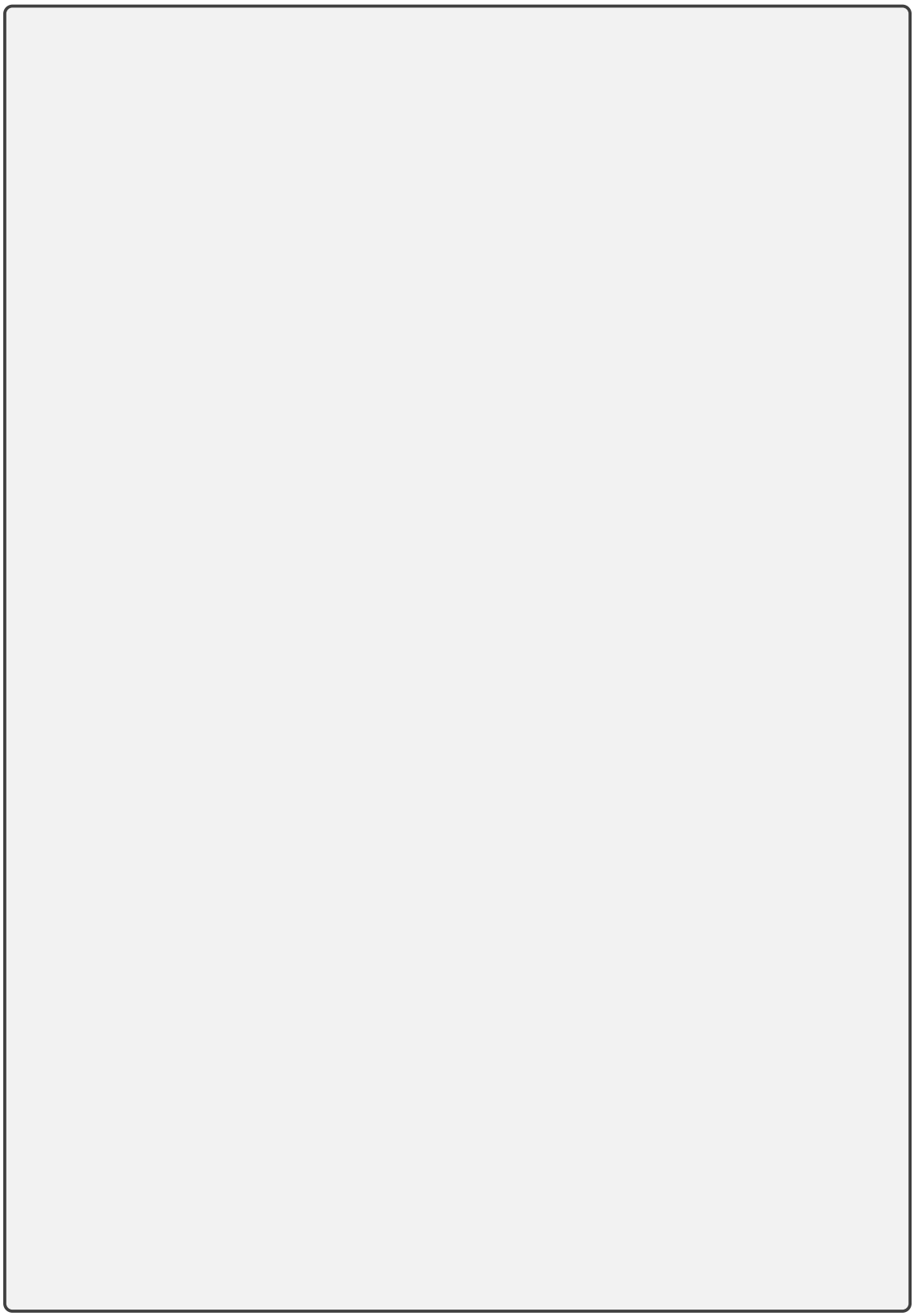


(3) [5 points] Write the Lagrangian for the R arm shown above.



(4) [5 points] Determine the Equations of Motion for the R arm shown above using the

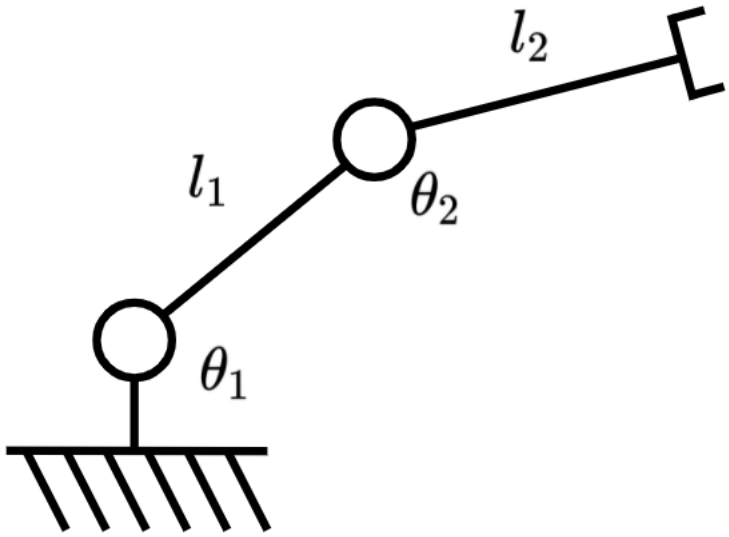
Lagrangian.



(5) [5 points] Rewrite the equations of motion into standard form.

1) RR robot

Consider the following robot




Given the above RR arm, each with a mass of $m_i$ located at $l_i/2$ and a moment of intertia of $I_i$ about their com

(1) [5 points] Determine the Kinetic Energy for the RR arm shown above.

(2) [5 points] Determine the Potential Energy for the RR arm shown above.

(3) [2 points] Write the Lagrangian for the RR arm shown above.

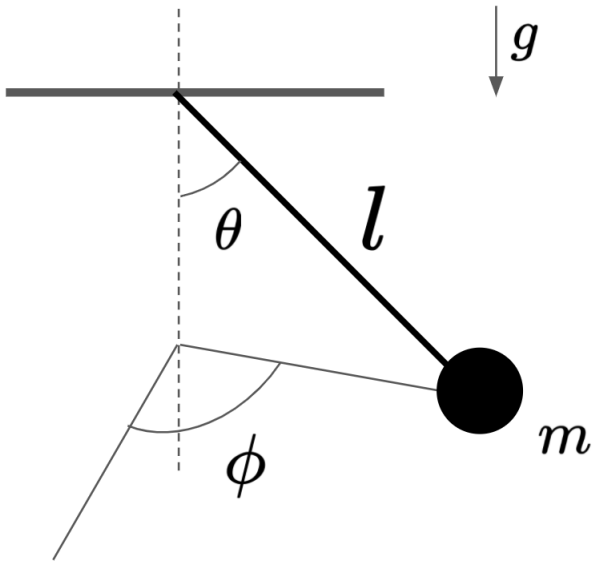
(4) [10 points] Determine the Equations of Motion with respect to $\theta_1$ for the RR arm shown above using the Lagrangian.

(5) [10 points] Determine the Equations of Motion with respect to $\theta_2$ for the RR arm shown above using the Lagrangian.

(6) [5 points] Rewrite the equations of motion into standard form.

2) 3D Pendulum

Consider the following robot




Given a pendulum that can move anywhere in 3D space, with a fixed length l and a mass m, solve for its equations of motion. Hint: use spherical coordinates.

(1) [5 points] Determine the Kinetic Energy for 3D pendulum.

(2) [5 points] Determine the Potential Energy for the 3D pendulum.

(3) [2 points] Write the Lagrangian for the 3D pendulum.

(4) [10 points] Determine the Equations of Motion with respect to $\theta$ for the 3D pendulum.

(5) [10 points] Determine the Equations of Motion with respect to $\phi$ for the 3D pendulum.

(6) [5 points] If $\phi$ is kept constant, what do the equations above reduce to? What system that we've studied in class has similar EOMs?

# 5 Code Questions

There is no coding portion in this homework.

# 6 Submission Checklist

- ☐ Create a PDF of your answers to the written questions with the name `writeup.pdf`.
- ☐ Upload `writeup.pdf` to Gradescope.